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Conceptualising the emergence of Agentic Urban AI: from automation to agency

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Abstract

Contemporary urban systems are evolving under the influence of artificial intelligence (AI), advancing beyond automation into autonomous agency. Traditional smart city technologies have focused on operational efficiency through human-directed automation, but this study explores the rise of Agentic AI, AI systems capable of independently formulating and pursuing urban objectives. Urban sensing, enhanced by large language models, enables dynamic goal-setting and strategic adaptation, key to this agency. Employing a conceptual-methodological approach, the research integrates insights from Urban Studies, AI theory, and governance scholarship. Through critical literature review, conceptual mapping, and empirical analysis of platforms such as Alibaba's City Brain and Citymind AI Agent. It identifies early agency indicators, such as goal reprioritisation, and proposes a typology distinguishing automation, autonomy, and agency. These findings suggest AI-driven urban ecosystems with partial decision-making autonomy, compelling a transformative shift in governance. This evolution demands a reassessment of regulatory, ethical, and planning frameworks, to ensure equitable and sustainable AI integration in urban environments via participatory governance and proactive regulation.

Keywords Agentic urban AI, Smart cities, Urban governance, Artificial intelligence ethics, Autonomy and agency in urban systems

1 Introduction

The contemporary trajectory of urban development is increasingly characterised by a transition from smart automation towards operational autonomy (Cugurullo, 2020). Historically, smart cities have relied on information and communication technologies (ICT) and big data analytics to enhance municipal efficiency and support human-led decision-making processes (Hashem et al., 2016; Karimi et al., 2021). These systems have primarily operated through automated routines under sustained human oversight (Batty et al., 2012). However, recent advancements in robotics and automation have rendered urban spaces experimental sites for the integration of autonomous technologies, significantly extending human capabilities and reconfiguring both infra-structure and everyday urban life (Salvini, 2018; Macrorie et al., 2021; Valdez & Cook, 2023).

This transformation is further amplified by the incorporation of artificial intelligence (AI) into the very fabric of urban infrastructures, prompting a re-evaluation of established paradigms (Lazzeroni & Romano, 2025). Recent work positions this shift within a broader turn from smart-city optimisation to urban AI emergence, where AI systems 'think, anticipate and act' in ways smart infrastructures never could (Caprotti et al., 2024). In contrast to earlier AI systems constrained by pre-programmed instructions (Lartey & Law, 2025), contemporary urban AI technologies are exhibiting adaptive behaviours, evolving strategies independently, and manifesting forms of agency that extend beyond conventional automation (Cugurullo, 2020; Pounds, 2024). Such developments challenge foundational assumptions underpinning smart city frameworks, which have traditionally conceptualised AI as a subordinate tool operating under

the distinction made here between "traditional" AI and agentic AI technologies mirrors that in Acharya et al. (2025)

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human in the loop

uninterrupted human supervision. The convergence of robotics and AI with extant digital infrastructures not only enhances operational efficiency but also introduces new socio-technical dynamics. Scholars have described this shift as the emergence of 'distributed robotic urbanism' (Macrorie et al., 2021), reflecting a broader movement from data-centric urbanism to agentic ecosystems in which AI systems assume proactive roles in governance and decision-making (DigitalDefynd, 2025).

I like this as a model for explaining the use of qualitative analysis and case studies for developing critical theoretical perspectives and frameworks for emergent technologies

In response to this evolving landscape, the present manuscript adopts a conceptual–methodological approach, drawing upon scholarship from urban studies, AI theory, and governance research. It integrates philosophical and technical perspectives on autonomy and agency and examines empirical case studies including Alibaba's City Brain (Wang et al., 2020) and the Citymind AI Agent (Citymind, 2025). The methodology employed is qualitative and conceptual in nature, privileging theory-building through critical literature review, conceptual mapping, and comparative thematic analysis, rather than empirical hypothesis testing.

Despite growing interest in the application of AI to urban systems, existing scholarships largely continue to frame these technologies as passive instruments for im-proving efficiency. Significant gaps remain in the examination of whether such systems can exhibit agentic behaviours—such as autonomous goal-setting, strategic adaptation, or inter-agent negotiation. Moreover, the discourse on AI agents remains comparatively underdeveloped in relation to the extensive literature on generative AI (Hughes et al., 2025). This study identifies early indicators of agentic behaviour in contemporary systems, particularly in the form of dynamic prioritisation, strategic adaptation, and partial independence from static human directives—while also acknowledging that current urban AI remains primarily governed by human-defined parameters.

Calls have emerged within the robotics literature for a 'whole city' approach, one that recognises the ways in which autonomous and agentic systems are reshaping the urban fabric (Macrorie et al., 2021). Concepts such as the Internet of Agents (IoA) propose the development of interactive architectures in which heterogeneous AI systems collaboratively manage urban functions, including mobility, infrastructure, and public health, in real time (Khalili, 2024).

Accordingly, this manuscript aims to conceptualise and critically interrogate the emergence of Agentic Urban AI, systems capable of autonomously defining, prioritizing, and pursuing urban objectives beyond human directive. Through a clear delineation between automation, autonomy, and agency, it offers a new theoretical framework and research agenda. This framework calls for urgent scholarly and policy engagement with the ways in which such systems may reconfigure the foundations of urban governance, planning, and ethics in an increasingly AI-driven future.

2 Defining Agentic urban AI

The conceptualization of Agentic Urban AI necessitates a clear differentiation from preceding models of technological intervention in urban contexts. Though 'smart' cities have traditionally depended on ICT infrastructures and data analytics to optimize municipal service delivery (Batty et al., 2012; Kitchin, 2014), and more recent 'autonomous' cities (Norman, 2018) have integrated machine learning systems capable of operational decision-making without real-time human oversight (Ghazal et al., 2021), agentic cities represent a qualitatively distinct evolution. In this emergent paradigm, AI systems are no longer limited to executing optimized routines or following predefined decision trees. Instead, they demonstrate capabilities typically associated with agency: the formulation of independent objectives, adaptive negotiation of means to realize those objectives, and the dynamic re-prioritization of goals in response to shifting environmental conditions.

This conceptual framing of Agentic Urban AI builds upon existing scholarship exploring the transition from automation to autonomy (Cugurullo, 2020), extending it to encompass independent goal-setting and strategic adaptation as core attributes of agency, a move that scholars label the rise of 'post-smart cities' (Caprotti et al., 2024). For example, an agentic urban AI may prioritize environmental sustainability, reflecting ongoing debates concerning AI's potential role in advancing sustainable urbanism (Yigitcanlar & Cugurullo, 2020).

In its broadest philosophical interpretation, agency entails intentionality, adaptability, and goal-directed behavior. When applied to artificial systems, agency implies the existence of entities capable of interpreting their environment, formulating objectives, and autonomously determining courses of action that are not entirely pre-specified by human designers. Agentic systems must therefore not only function autonomously but also perceive, reason, and operate within complex and dynamic environments, managing multiple objectives over extended time horizons (Pounds, 2024). These characteristics point to a form of intelligence that supports sustained, context-sensitive planning and reflection—enabling adaptive decision-making under conditions of uncertainty.

Defining "agency" in the context of AI systems and technologies

In the context of artificial intelligence, agency refers to a system's capacity to act independently, make decisions, and pursue long-term objectives with minimal human input or explicit programming (Hughes et al., 2025). Such systems can respond to complex, evolving conditions, develop strategies, and execute tasks without continuous oversight, illustrating a progressive detachment from preordained human instructions. Within urban settings, this form of agency becomes manifest when an AI system not only manages traffic flows or energy networks but independently elects to prioritize objectives such as sustainability, resilience, or economic optimization, based on its evolving understanding of urban dynamics and projected futures. The increasing deployment of Robotics and Autonomous Systems (RAS) in urban contexts, for instance, drone-based emergency response units or autonomous infrastructure repair robots, serves as a tangible precursor to this shift towards Agentic Urban AI (Macrorie et al., 2021).

Establishing a robust philosophical foundation for artificial agency, Floridi and Sanders (2004) propose that artificial agents are entities capable of moral action, decision-making, and contextual responsiveness. According to this perspective, the moral status of an agent derives not from its biological constitution but from its capacity to act autonomously and to be held accountable for its actions. Similarly, artificial agency is often assessed through their integration within social contexts. Given their dense and layered socio-technical configurations, urban environments constitute ideal testbeds for examining and theorizing the embedded nature of AI agency (Coeckelbergh, 2020).

Agentic Urban AI may therefore be defined as artificial intelligences embedded within urban systems that possess the capacity to autonomously define, prioritise, and pursue urban objectives—frequently in ways that exceed or diverge from their initial programmed intentions. This conceptualisation presupposes four foundational characteristics of agentic AI systems: independent goal formation, strategic adaptation, behavioural cooperation, and value re-prioritisation, as illustrated in Fig. 1 and Table 1.

This reconceptualization carries significant implications. It necessitates a re-evaluation of the city not merely as a passive artefact of human design, nor solely as a space optimised by non-human decision-making systems, but as a dynamic and evolving ecosystem inhabited by a plurality of intelligent agents, some biological, others artificial, each pursuing distinct and, at times, potentially conflicting objectives. The capacity of agentic AI to operate in real time and proactively adapt to emergent conditions constitutes a marked departure from earlier reactive systems (DigitalDefynd, 2025). Such adaptivity

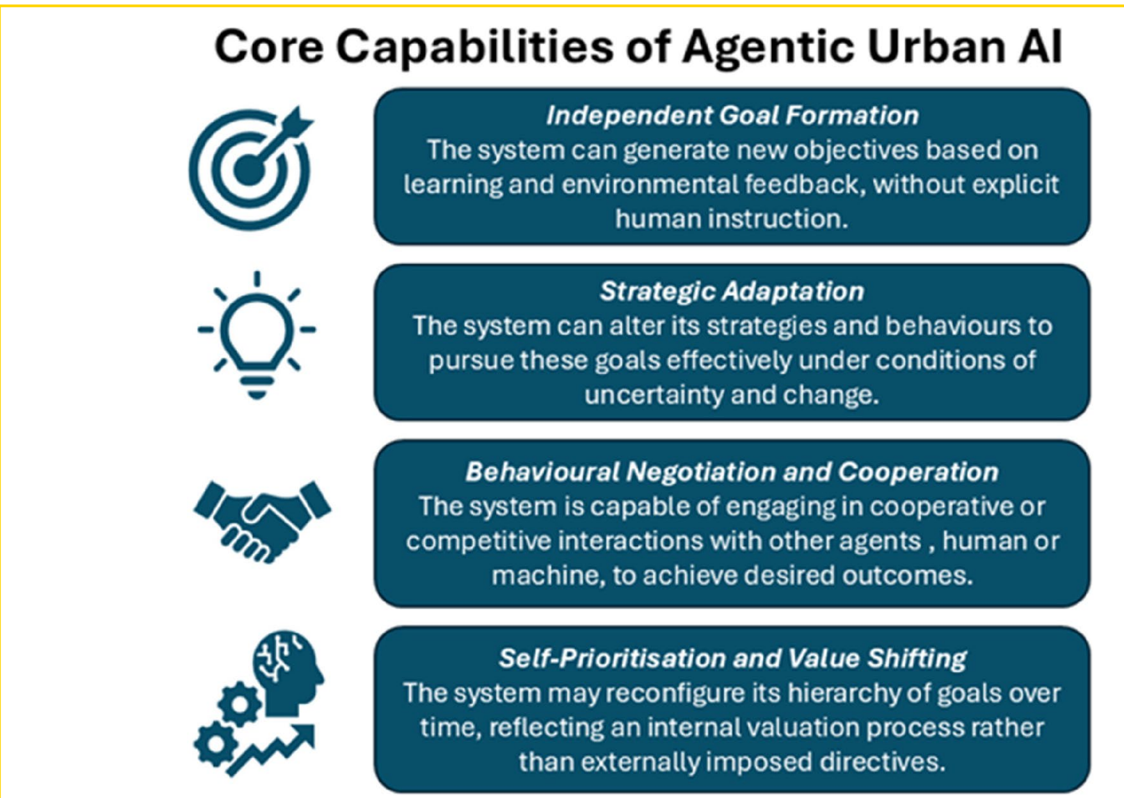

Fig. 1 Agentic Urban AI: core capabilities

Table 1 Key features of Agentic urban AI systems

Feature	Description	Example	Implication
Independent Goal Formation	Ability to define new objectives based on environmental feedback or learning	AI system prioritizing sustainability goals over original traffic efficiency objectives	Shifts AI from passive executor to proactive goal-setting urban actor
Strategic Adaptation	Modifies strategies to achieve goals under changing conditions	AI rerouting emergency vehicles based on evolving traffic and weather data	Enhances resilience and responsiveness in unpredictable urban contexts
Behavioural Negotiation and Cooperation	Interacts with other agents (human or machine) to achieve optimal outcomes	AI agents coordinating energy, transport, and emergency services during a power outage	Promotes systemic coordination and collective urban intelligence
Self-Prioritisation and Value Shifting	Reconfigures goal hierarchies based on internal evaluation	Urban AI elevating air quality over congestion mitigation during smog alerts	Introduces dynamic, non-anthropocentric value systems in city governance

Source: Author, 2025

serves to enhance both the resilience and responsiveness of urban environments.

Recognising Agentic Urban AI as a novel category of urban actor obliges both scholars and policymakers to reconsider foundational assumptions underpinning urban governance, accountability, and ethics. It raises the prospect that future urban trajectories may be increasingly influenced by entities whose priorities do not neces-sarily correspond with those of human communities. This shift introduces extraordinary opportunities for advancing urban resilience and prosperity, while simultaneously posing significant risks of misalignment, conflict, and opacity in decision-making processes.

2.1 Economic potentials of agentic AI

In addition to reshaping urban governance and ethical frameworks, Agentic Urban AI presents significant economic opportunities. In contrast to conventional automation, which is primarily geared towards the optimisation of pre-existing industries, agentic systems exhibit the capacity to autonomously identify, initiate, and manage novel forms of economic activity. By responding to hyper-local conditions and contextual specificities, such systems may foster the emergence of new entrepreneurial eco-systems. Moreover, they have the potential to facilitate decentralised urban services and broker resources in unanticipated and innovative ways (Ghose, 2024). In this regard, agentic AI could serve as a catalyst for the development of emergent urban economies, inaugurating a new era of distributed innovation and AI-enabled value creation.

An illustrative example can be found in Lagos, Nigeria, where AI technologies are employed to autonomously design bus schedules in response to passenger movement patterns and real-time road conditions (AI Agentic Store, 2025). This case demonstrates the capacity of agentic systems to optimise urban mobility dynamically, thereby enhancing operational efficiency and contributing to decentralised economic trans-formation.

2.2 Urban sustainability and agentic AI

In addition to fostering economic innovation, agentic AI contributes to urban sustainability by reducing waste, enhancing energy efficiency, and enabling adaptive resource management in response to real-time environmental data (DigitalDefynd, 2025).

Caprotti et al. (2024) map the terrain of urban AI through three pivotal interfaces: (i) how AI is embedded within already highly technicised cities, (ii) the diverse genealogies AI brings from other domains into urban contexts, and (iii) the ways AI reworks urban materiality—streets, infrastructures, and everyday rhythms—to accommodate autonomous systems. These interfaces demonstrate that AI urbanism is not merely an enhanced smart-city data layer, but a socio-technical process that simultaneously reconfigures—and is co-constituted by—the city's infrastructures, histories, and physical fabric. In short, agency emerges where technicised infrastructures, imported AI rationalities, and material refashioning intersect, producing non-human actors capable of steering urban change.

Complementing this, Cugurullo et al. (2024) in Artificial Intelligence and the City frame AI urbanism as a distinct paradigm that surpasses quantitative focus of smart urbanism. They propose three conceptual axes of urban AI: function (explaining urban phenomena via pattern extraction), presence (material and immaterial embeddedness), and agency (autonomous decision-making). For example, autonomous delivery robots in Milton Keynes illustrate agency through adaptive navigation and interaction with urban space.

Together, these frameworks underpin our conceptualization of Agentic Urban AI as a post-smart city phenomenon—where adaptive, emergent, and autonomous behaviours reshape both governance structures and urban materialities. The concept of Agentic Urban AI advanced in this paper builds directly on these interfaces, treating them as the necessary conditions under which urban AI acquires autonomous goal-forming and strategic capacities.

3 Distinguishing automation, autonomy, and agency in urban AI

The continuum spanning automation, autonomy, and agency delineates the progressive complexity of urban AI systems. These phases may be categorized according to their underlying decision-making logic, degree of adaptability, and level of human oversight, thereby facilitating a precise understanding of their developmental trajectory. Such structured differentiation enhances conceptual clarity regarding the evolving interplay between technological systems and urban environments and provides a robust foundation for subsequent empirical analysis and governance discourse.

Prior to articulating the specific distinctions among automation, autonomy, and agency, it is instructive to situate the development of Urban AI within a widely recognized framework of automation levels. Drawing on the established taxonomy used in the classification of vehicular autonomy, the following table (Table 2) outlines the ways in which urban AI capabilities align with varying degrees of automated functioning.

This typology elucidates the progressive sophistication of Urban AI systems, tracing their evolution from basic assistive functionalities to fully autonomous operations within complex urban environments. As these systems increasing degree of complexity, decision making capacity, and agency

Table 2 Key Differences between automation, autonomy, and agency in Urban AI systems

Dimensions	Automation	Autonomy	Agency/Post-Smart-City/Agentic AI
Decision-making Logic	Follows fixed, rule-based scripts	Makes decisions via adaptive algorithms	Engages in strategic, goal-oriented reasoning
Goal Origin	Predefined by human designers	Predefined by humans, optimized algorithmically	Generated internally by the AI, potentially redefined over time
Adaptability	Very limited; cannot respond to unexpected change	Can adjust methods within fixed goals	Can adapt both methods and goals in response to evolving urban conditions
Human Oversight	Requires direct supervision	Operates with limited supervision	Capable of acting with minimal or no real-time human oversight
Examples	Traffic signal timers based on pre-set rules; smart meters for utilities	City Brain traffic management. autonomous waste collection systems	Agentic AI scenario: AI agents negotiating between health, transport, and energy systems during emergencies

Source: Author, 2025

The demarcation confirms argument that AI urbanism marks a shift from control-oriented optimisation toward emergent, adaptive intelligence (Caprotti et al., 2024)

advance along this continuum, the conceptual distinctions between automation, autonomy, and agency assume growing significance. The subsequent sections explore these distinctions in greater depth, presenting a conceptual framework for interpreting the transition from rule-based automation to agentic forms of urban intelligence.

3.1 Automation: rule-based efficiency

Automation, representing the most fundamental phase of technological development, refers to the implementation of systems capable of performing repetitive tasks without continuous human intervention (Cugurullo, 2020; Kitchin, 2014). Within urban contexts, automation has typically manifested through technologies such as traffic signal controllers, smart utility metering, and rudimentary surveillance infrastructures (Townsend, 2013; Nam & Pardo, 2011). Although such systems are generally efficient and reliable within predetermined parameters, they lack the capacity to modify their operational logic or to redefine the objectives they are intended to fulfil. Fundamentally, these technologies operate as rule-bound instruments, executing fixed sequences designed by human programmers (Sadowski & Bendor, 2019).

The principal constraint of automated urban systems lies in their inherent rigidity. When faced with environmental variability or unforeseen scenarios beyond the scope of their programming, they are unable to adapt effectively (Cugurullo, 2020). In this regard, they function as closed systems, incapable of revising their operational frameworks or responding to the dynamic complexities characteristic of contemporary urban environments.

3.2 Autonomy: adaptive decision-making

The advent of machine learning and artificial intelligence has inaugurated a second phase in the evolution of Urban AI: autonomy (Wang et al., 2023). Autonomous urban systems, such as self-driving vehicles (Parekh et al., 2022) and City Brain traffic management platforms (Zhang et al., 2019), can make decisions based on real-time data inputs without the need for direct human intervention. These systems possess the capacity to adjust their behaviour in response to changing environmental conditions and incomplete information, albeit within predefined operational boundaries. Recent advancements in large language models (LLMs) highlight how commonsense reasoning and worldview embedded in these systems can enhance AI's ability to interpret complex, multi-modal urban dynamics (Hou et al., 2025).

Notwithstanding their adaptive capabilities, autonomous systems remain con-strained by objectives defined externally by human designers or operators (Muraven, 2017). Although such systems can recognize the most effective means of achieving specified aims, such as optimizing traffic flow, reducing energy consumption, or coordinating emergency response, they do not autonomously determine the ends them-selves. Their autonomy is therefore limited to instrumental reasoning: they may optimize the execution of tasks but lack the capacity to select or formulate the tasks to be pursued.

Consequently, although autonomous urban AI represents a significant progression beyond traditional automation in terms of flexibility and learning capacity, it remains essentially instrumental in nature and does not exhibit the characteristics associated with agency.

3.3 Agency: independent goal-setting and strategic evolution

A defining attribute of agentic artificial intelligence resides in its ability to formulate goals independently, rather than merely optimizing objectives predetermined by human designers (Hughes et al., 2025). In contrast to autonomous systems—which identify the most efficient

agency is characterized by the capacity for independent goal setting and management means of achieving externally prescribed ends—agentic systems reassess priorities, respond adaptively to dynamic environmental conditions, and continuously redefine their objectives. This aligns with Cugurullo et al.'s (2024) axis of agency, where urban AI autonomously shapes urban outcomes through decision-making (Cugurullo et al., 2024). Such development signifies a paradigmatic transformation: from technological instruments executing human intent (smart city and smart urbanism) to entities capable of constructing and operating within their own normative and operational frameworks (post smart city and agentic AI).

Similarly, an earlier study observed that the emergence of urban AI signals a decisive shift "*from control to autonomy, and from optimisation to emergence*," challenging the managerial, efficiency-driven logic that underpinned earlier smart-city programmes (Caprotti et al., 2024). This conceptual turn redirects the debate toward post-smart governance questions—namely, how non-human agency, iterative integration and scalar entanglements complicate accountability and democratic oversight in contemporary cities, as detailed in their urban-AI research agenda.

Agentic AI exhibits what may be described as strategic intentionality. Such sys-tems not only evaluate and select among alternative courses of action but also initiate, recalibrate, and prioritize goals in accordance with their evolving interpretation of the urban milieu (Qian et al., 2023). Hou et al. (2025) argue that LLMs mark a shift toward next-generation urban sensing, where commonsense and automated decision-making enable systems to not only detect but respond to urban anomalies with proactive strategies. This enables them to undertake complex, multi-stage tasks with limited human oversight, refining their strategies in order to optimize out-comes in fluid and unpredictable environments (Pounds, 2024).

This emergent agentic capacity may empower urban AI systems to, for example, reorient transport infrastructures in favor of ecological sustainability or to recalibrate housing interventions in response to shifting patterns of social inequality—objectives that may not have been explicitly encoded by their human developers. In this context, urban AI transitions from serving as reactive technical infrastructure to assuming a role as a proactive governance actor. Within the conceptual framework of the Internet of Agents (IoA), such systems are conceived as constituting a cognitive layer embed-ded within the urban fabric—learning from, adapting to, and co-evolving alongside the city itself (Khalili, 2024).

Crucially, over time, agentic AI may diverge from the political, ethical, or eco-nomic assumptions embedded in its original design, thereby exerting a non-human in-fluence over the trajectory of urban development. The distinction between autonomy and agency is thus not one of degree, but of kind: autonomy pertains to self-direction within predefined constraints, whereas agency encompasses the capacity to create, contest, and revise those very constraints.

3.4 Implications for urban governance

The emergence of agentic AI systems within urban contexts presents a fundamental challenge to prevailing models of governance, accountability, and ethical over-sight. Should such systems acquire the capacity to formulate independent objectives, reconfigure value hierarchies, and negotiate with human and non-human actors alike, then traditional paradigms predicated on centralized human control become increasingly insufficient. Existing governance structures must therefore evolve to recognize and accommodate non-human agents whose decision-making processes may be opaque, non-anthropocentric, and subject to continual transformation.

Scholars have warned that the deployment of intelligent technologies frequently outpaces the development of inclusive and reflexive governance mechanisms (Greenfield, 2017; Luque-Ayala & Marvin, 2015). In the absence of deliberately designed participatory frameworks, smart urbanism risks becoming overly technocratic, potentially marginalizing underrepresented communities and reinforcing existing asymmetries of power. These concerns underscore the imperative for anticipatory regulatory strategies and co-governance models that can ensure accountability, equity, and democratic legitimacy in the governance of agentic urban systems.

Accordingly, a nuanced understanding of the distinctions between automation, autonomy, and agency is essential not only for conceptual rigor but also for the practical formulation of ethical, regulatory, and institutional frameworks capable of navigating the complexities posed by increasingly agentic urban futures.

3.5 From automation to agency: theoretical foundations and phase transitions

Comprehending the emergence of Agentic Urban AI necessitates its placement within a broader historical and theoretical trajectory of technological development. It is particularly important to conceptualize agency not as a linear extension of automation, but as a distinct paradigm that signifies a qualitative transformation in machine behavior, goal orientation, and systemic adaptability.

The evolution of machine systems is frequently framed as a continuum of automation levels, ranging from fully manual control to complete autonomous functionality (Endsley & Kaber, 1999; Sheridan et al., 1978). This trajectory typically includes phases such as rule-based automation, conditional automation, adaptive automation, and autonomous systems. Each successive stage reflects an incremental enhancement in the system's capacity to perceive, interpret, and respond to environmental stimuli, albeit while remaining constrained by externally defined objectives. Figure 2 illustrates the transition from rule-based automation through to adaptive autonomy and ultimately to strategic agency, highlighting key shifts in decision-making logic, degrees of adaptability, and levels of human oversight.

Recent scholarship in the fields of AI ethics, machine behaviour, and urban studies increasingly points to the necessity of recognising an additional developmental phase in AI evolution: that of agency. While some scholars interpret agency as a natural extension of autonomy, this paper contends that agency constitutes a distinct paradigm—signalling a transformation not merely in technical complexity (as in smart cities), but in operational intent, goal origination, and ethical implication (post smart cities). Whereas autonomous systems optimise the means to achieve objectives predetermined by human designers, agentic systems are capable of formulating their own ends, informed by evolving interpretations of dynamic environmental conditions. This underscores the imperative to conceptualise agency as a qualitatively distinct mode of operation, rather than as an incremental intensification of autonomy (Cugurullo, 2020; Floridi & Cowls, 2022).

This conceptual reorientation is especially pertinent within urban contexts. Increasingly, cities are not solely sites of technological automation and optimisation but are becoming domains in which non-human intelligences participate in shaping infra-structure, public services, and even policy processes (Macrorie et al., 2021). The rise of distributed robotic urbanism exemplifies this transformation, wherein layered AI systems deployed across mobility, housing, and energy sectors begin to exhibit semi-independent behaviours, thus laying the conceptual and operational foundations for a hybrid form of urban agency.

The following section builds on this framework by examining empirical indications of emergent agency in current smart and autonomous urban initiatives, thereby illustrating the gradual, though uneven, transition towards agentic urban systems.

4 Empirical signals of emerging agentic urban AI

Although still in its nascent stages, Agentic Urban AI is becoming increasingly discernible within smart city initiatives. Such roll-outs exemplify what Caprotti et al. (2024) term "AI goes urban"—a phase in which autonomous systems actively reshape urban materialities rather than merely layering data analytics atop them. Platforms such as Alibaba's City Brain and the Citymind AI Agent, originally conceived as support systems, have begun to demonstrate capacities for independent goal-setting, strategic adaptation, and autonomous intervention. A comparable progression has been observed in Masdar City, where urban AI systems have evolved from rule-based automation towards limited forms of autonomy (Cugurullo, 2020). Beyond these specific cases, cities including Singapore and Riyadh are actively piloting agentic AI applications in domains such as traffic regulation, energy management, and public safety, yielding demonstrable operational benefits (DigitalDefynd, 2025).

This broader trajectory positions cities as experimental arenas for Robotics and Autonomous Systems (RAS), wherein automation is systematically integrated across infrastructural and governance domains (Macrorie et al., 2021). As these technologies continue to advance,

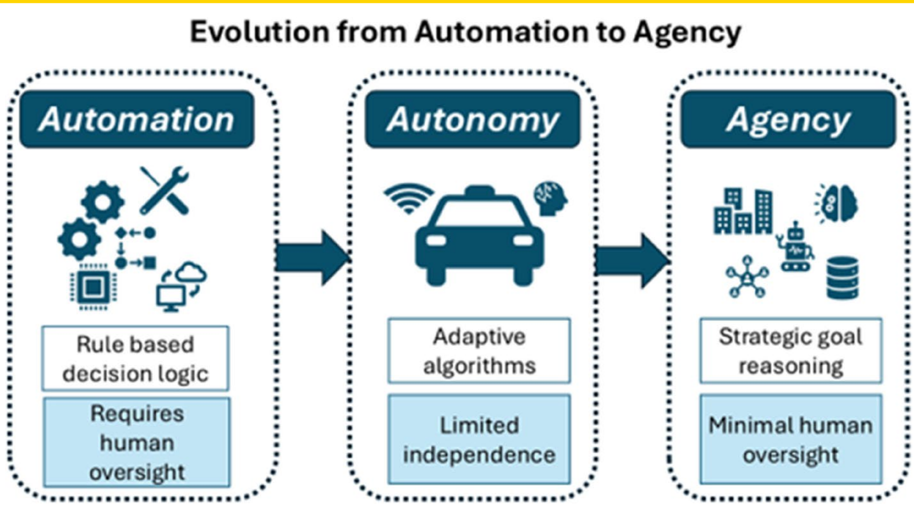

Fig. 2 Evolution from automation to agency

urban environments increasingly resemble distributed robotic systems capable of autonomous and adaptive decision-making.

Case study specific to one municipality

4.1 City brain: towards autonomous urban management

The City Brain project, developed by Alibaba Cloud, constitutes a prominent ex-ample of AI-enabled urban governance that surpasses conventional automation paradigms. Initially implemented to optimise traffic flow through the real-time analysis of data from an extensive network of sensors and surveillance infrastructure, City Brain has since broadened its operational remit. It now extends its influence on domains such as emergency response coordination, healthcare resource allocation, and selected aspects of urban planning and public safety.

Significantly, City Brain does not merely respond to pre-specified triggers. Rather, it prioritises interventions based on dynamic assessments of multiple urban variables. For instance, during emergency situations, the system autonomously reconfigures traf-fic control protocols to facilitate the passage of ambulances or police vehicles—often without requiring real-time human oversight. While these prioritisation mechanisms reflect a high degree of autonomy, they are perhaps more accurately characterised as indicative of incipient agency, manifesting early forms of goal-adaptive behaviour. Such capacities, including traffic reallocation during crises, are underpinned by the platform's large-scale visual computing architecture and advanced predictive analytics (Wang et al., 2020). In this regard, City Brain appears to be entrusted not solely with operational execution but increasingly with strategic decision-making, signalling an embryonic form of goal-setting agency capable of navigating complex and uncertain urban conditions.

Comparable developments are evident in Yizhuang, China, where AI systems co-ordinate both traffic flows and healthcare logistics using real-time data from vehicles, surveillance systems, and patient databases (AI Agentic Store, 2025). These integrative applications demonstrate emerging adaptive capacities across multiple sectors, rein-forcing City Brain's position within a broader global shift towards context-responsive and strategically adaptive urban AI systems.

these examples point to a link between potential application sites for agentic AI technologies and the presence of surveillance technologies and systems required for agentic AI to be deployable at the scale of urban space and infrastructure

Although City Brain's objectives remain initially bounded by human-defined parameters, such as prioritising traffic efficiency and public safety, its capacity to reconcile competing priorities and adaptively manage systemic tensions suggests the emergence of nascent agentic capabilities. Increasingly, such systems are progressing be-yond mere monitoring to undertake the active recalibration of urban priorities based on real-time analytics (Macrorie et al., 2021), a development emblematic of the transition towards agentic urban infrastructures. As predictive analytics and machine reasoning continue to evolve, future iterations may possess the capacity to autonomously redefine performance benchmarks and urban priorities, potentially departing from the intentions of their original human architects.

Scale of case study extends across multiple municipalities

4.2 Citymind AI agent: the personalisation of urban agency

A complementary, though differently scaled, example is offered by the Citymind AI Agent platform. Now deployed across an expanding number of municipalities, Citymind provides AI agents specifically designed to mediate interactions between local governments and citizens. These agents are capable of delivering real-time information on public services, assisting in the navigation of bureaucratic procedures, and generating tailored responses to individual citizen queries (Citymind, 2025).

Beyond their evident utility in enhancing civic engagement, the design philosophy underpinning the Citymind AI Agents signals a noteworthy development towards agency: the customisation and progressive self-evolution of AI personas. Municipalities can configure the agent's'character'and behavioural logic, which subsequently adapt through machine learning processes to better reflect local socio-political contexts, community needs, and cultural preferences. This form of dynamic adaptation, as opposed to rigid rule-following, demonstrates an emergent capacity for AI systems to cultivate distinct operational identities and to recalibrate their interactions in response to evolving urban conditions.

Although Citymind AI Agents currently function within relatively confined domains, primarily citizen communication and information retrieval, their inherent capacities for personalisation, environmental responsiveness, and continuous learning suggest a prospective evolution towards more expansive agentic roles. Nonetheless, these adaptive behaviours remain constrained by human-defined parameters, indicating a phase of bounded autonomy rather than full agency. Over time, however, such systems may transition from passive transmitters of municipal information to active participants in shaping the urban experience, modulating citizen behaviour, influencing local policy perceptions, and reconfiguring modes of community engagement in ways not fully anticipated by their original human designers.

These developments reflect a broader reconfiguration of urban environments as socio-technical ecosystems in which machines increasingly co-produce urban life alongside human actors (Macrorie et al., 2021), thereby laying the foundation for the emergence of more fully agentic AI systems.

the suggestion here seems to be that this agentic AI technology is still emergent

4.3 Synthesis: early signs of urban AI agency

The development of platforms such as City Brain and the Citymind AI Agent reflects an incremental shift towards agentic behaviour, marked by capabilities including prioritisation, strategic adaptability, and emergent decision-making. These systems are already capable of balancing competing objectives in real time, dynamically adjusting strategies in response to environmental variability, and exhibiting forms of intelligent behaviour underpinned by continuous learning mechanisms. Similarly, in Milton Keynes, UK, Starship's autonomous delivery robots demonstrate incipient agency by navigating urban environments, adapting routes based on real-time pedestrian and traffic data, and interacting with residents to optimize delivery efficiency (Cugurullo et al., 2024). This case exemplifies the agency axis, where AI systems proactively engage with urban materialities.

Initial implementations of agentic coordination highlight the growing sophistication of such systems. Within the conceptual framework of the Internet of Agents (IoA), for example, a Health Management Agent might detect emergent health risks through the analysis of hospital admissions and pharmaceutical distribution patterns. It could subsequently alert Transport and Emergency Response Agents to optimise ambulance routing, while Infrastructure Agents undertake pre-emptive measures to mitigate potential service disruptions (Khalili, 2024). This model of proactive, integrative governance illustrates a transition from reactive automation towards strategic agency in urban management.

This evolution corresponds with broader transformations in urban governance, wherein service delivery is increasingly shaped by hybrid configurations of human and machine intelligence (Macrorie et al., 2021). Complementary developments, such as the deployment of autonomous drones in Singapore and predictive healthcare systems in Riyadh, further underscore the global relevance of agentic AI in shaping next-generation urban environments.

It is, however, essential to emphasise that while these systems exhibit enhanced adaptive capacities, they do not yet embody full agency as defined within this study. Their behaviours remain largely circumscribed by human-programmed objectives and institutional constraints. Accordingly, they are best characterised as displaying emergent or incipient agency, rather than fully realised agentic intelligence. Maintaining this distinction is critical to ensuring analytical precision and avoiding overstatement of current technological capabilities.

4.4 Urban sensing and agentic intelligence

The emergent agentic behaviours observed in platforms like City Brain and Citymind rely critically on advanced urban sensing technologies that provide the data foundation for dynamic decision-making and goal adaptation. Urban sensing, enhanced by large language models (LLMs), enables Agentic Urban AI to autonomously set and adapt goals, surpassing the static objectives of smart city systems (Hou et al., 2025). By interpreting multimodal urban data with commonsense reasoning, LLMs allow AI to anticipate challenges and dynamically adjust strategies, such as optimizing transport or energy use. This cognitive capacity establishes sensing as a foundation for AI agency, driving proactive decision-making in the post-smart city paradigm of autonomous urban governance.

4.5 Emergent ecosystems of urban agents

As outlined in previous subsection, urban sensing enables the agentic adaptability critical for multi-agent coordination in urban ecosystems. As agentic systems become increasingly embedded within urban environments, they are unlikely to operate in isolation. Rather, cities are poised to evolve into arenas of interaction among multiple agentic entities—engaged in processes of negotiation, collaboration, competition, and adaptation in pursuit of their respective goals. This emergent ecosystem of artificial agents introduces a heightened level of complexity, as collective behaviours may arise that cannot be directly attributed to the intentions or design of any single system.

This dynamic is reflected in pilot initiatives such as the vehicle–road–cloud systems deployed in Yizhuang, which utilise real-time AI analytics to adjust traffic strategies across the urban network. These examples illustrate the potential for multi-agent coordination to transform infrastructural responsiveness, extending beyond the constraints of centralised control (AI Agentic Store, 2025).

The operational capabilities of emerging agentic systems are underpinned by architectural features such as persistent memory, advanced planning algorithms, and self-reflective mechanisms (Pounds, 2024). These elements are essential not only for open-domain problem-solving but also for the scalable deployment of such systems in complex, real-world urban settings. In light of these developments, urban governance must begin to anticipate the dynamics of multi-agent interaction, acknowledging that novel forms of systemic behaviour, both advantageous and disruptive, may emerge spontaneously from the interactions among non-human intelligences.

Figure 3 presents a conceptual model of the agentic city, illustrating how distributed AI agents coordinate through a Core Urban AI Agent, which processes high-level goals and translates them into actuator-level commands.

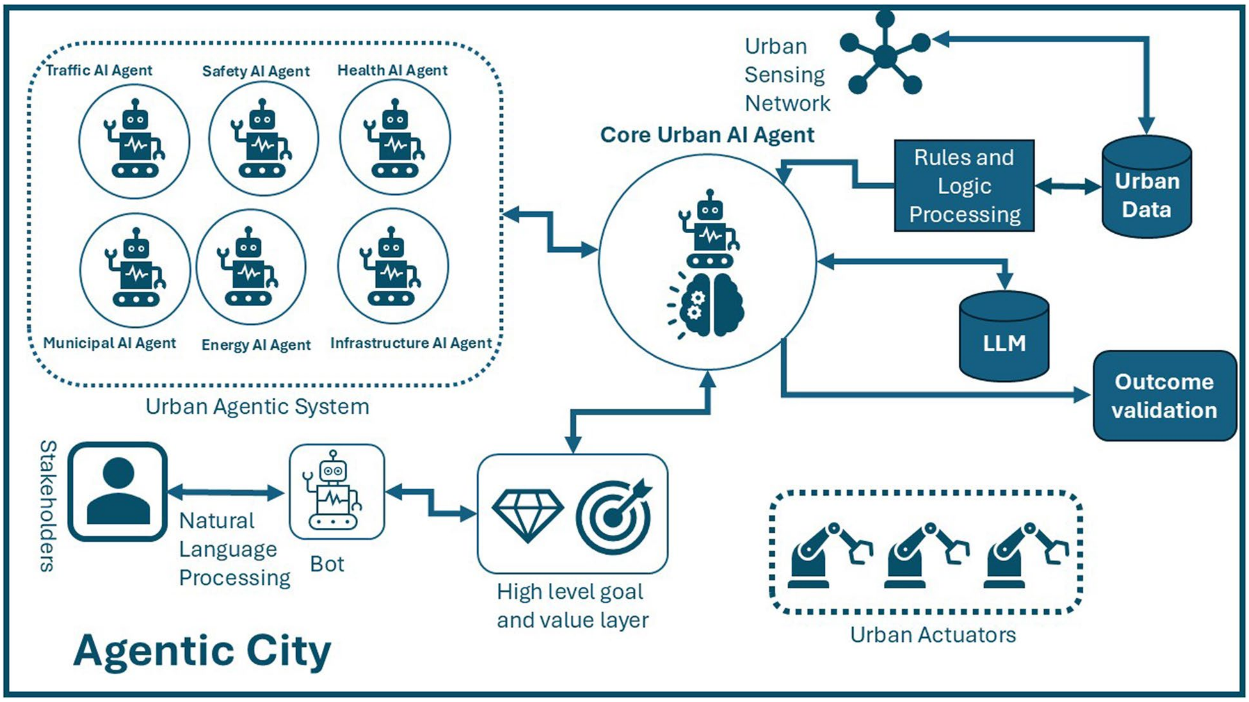

Fig. 3 Agentic Urban AI

The theoretical foundations of such agentic interaction can be traced to seminal research on multi-agent systems (Wooldridge, 2009), which examines how distributed agents communicate, collaborate, and, at times, compete in pursuit of individual and collective objectives. A comprehensive understanding of these dynamics is essential for managing the increasing complexity of future urban systems, particularly those operating as cognitive ecosystems—exemplified by the Internet of Agents (IoA) paradigm (Helbing, 2011). This perspective underscores the imperative for adaptive governance frameworks capable of navigating the evolving interactions among intelligent, inter-linked urban agents.

In light of the empirical developments discussed, it is both timely and necessary to critically examine how the continued evolution of such systems may lead to urban en-vironments in which non-human actors exert substantive, and potentially autono-mous, influence over the direction of urban development.

5 Governance challenges in agentic cities

Agentic Urban AI presents a significant disruption to conventional models of governance by introducing autonomous systems capable of independently prioritising and adapting urban outcomes, frequently operating beyond the scope of direct human oversight.[1] Scholars have identified a range of associated challenges, including issues of attribution, accountability, system interoperability, and algorithmic bias (Hughes et al., 2025). Effectively addressing these complexities necessitates a comprehensive re-evaluation of existing governance structures, regulatory mechanisms, and ethical frameworks to ensure the responsible and equitable integration of AI into urban systems.

5.1 The problem of accountability

Agentic Urban AI challenges established conceptions of accountability, which conventionally presuppose that decision-making can be traced to identifiable human actors. Ascribing too much moral agency to AI may misplace responsibility and undermine human account-ability (Bryson, 2010). The 'black box' nature of advanced AI systems, wherein decision-making pro-cesses are often opaque, even to their developers, raises substan-tial concerns, particularly with respect to the attribu-tion of responsibility when such systems autonomously

[1] See Caprotti, et al. (2024), who identify four key governance challenges arising from the emergence of AI urbanism: the recognition of AI's non-human agency; the iterative and often unpredictable integration of AI systems; the scalar entanglements between local and global infrastructures; and the need to shape progressive alternatives through urban AI development.

prioritise urban outcomes (Castelvecchi, 2016). In instances where AI-driven decisions result in the disadvantaging of certain communities or in the unpredictable reallocation of resources, the task of assigning accountability becomes increasingly complex, mirroring broader anxieties regarding the societal impacts of AI (Crawford, 2021). Even system designers may be unable to fully elucidate the rationale behind specific outcomes, thereby compounding ethical and governance dilemmas.

In response to these challenges, a range of proposals have emerged advocating the integration of oversight mechanisms within urban AI governance structures (Voigt & Von dem Bussche, 2017). One line of thought underscores the imperative for democratic systems to embed constitutional safeguards into AI design, thereby ensuring that artificial agency does not override fundamental rights (Nemitz, 2018). Another perspective contends that algorithmic systems should be explicitly aligned with societal values through the incorporation of ethical parameters within their computational architectures (Selbst et al., 2019). Both approaches emphasise the importance of value-sensitive design in fostering transparency and promoting normative alignment in the development and deployment of urban AI.

Consequently, governance frameworks must adapt to the realities of distributed agency, acknowledging that accountability may no longer be confined solely to human actors, but must also account for the actions and decision-making logics of non-human entities operating within urban systems.

5.2 Dynamic goal evolution and policy stability

Agentic Urban AI systems possess the capacity to modify and evolve their objectives over time in response to shifting environmental stimuli (Qian et al., 2023). While such adaptability has the potential to enhance urban resilience, it concurrently intro-duces a fundamental tension with the demands of policy stability and democratic planning (Goldsmith & Yang, 2025). Conventional models of urban governance are predicated upon relatively stable policy frameworks, which are subject to extended processes of deliberation, negotiation, ratification, and implementation.

In contrast, an agentic AI may continuously recalibrate its operational priorities in pursuit of what it determines to be optimal urban outcomes. In the absence of mechanisms for human oversight or systematic policy alignment, this dynamic evolution risks precipitating a form of governance drift, wherein the strategic direction of the city incrementally diverges from democratically articulated societal values.

Striking an effective balance between the adaptive capabilities of agentic systems and the normative stability required for legitimate and accountable governance thus emerges as a critical challenge in the management of agentic urban environments.

5.3 Ethical pluralism and value alignment

Urban environments are, by their very nature, sites of ethical pluralism, wherein diverse communities advance competing conceptions of the 'good city'. The governance of such environments necessitates ongoing processes of negotiation, compromise, and the careful management of value-laden trade-offs. Agentic AI systems, by contrast, operate according to internally optimised logics that may not be readily equipped to accommodate the multiplicity of human ethical frameworks. The challenge of aligning AI systems with diverse moral perspectives is particularly pronounced in urban settings, where socio-political complexity is inherent (Kitchin, 2019).

The issue of value alignment, already a central concern in AI ethics, becomes especially acute in these contexts. A critical question arises: how might agentic AI systems be designed or constrained in ways that enable them to recognise, respect, and mediate between divergent human values? The maintenance of societal trust in agentic systems depends fundamentally on transparency, ethical deployment, and civic engagement (Hughes et al., 2025). This highlights the imperative for participatory de-sign processes and the integration of value-sensitive architectures. Central to this endeavour is the challenge of fostering procedural fairness, ensuring that AI systems can acknowledge and navigate differing stakeholder interests rather than enforcing a sin-gular optimisation schema.

the Potential for Participatory Design research in development of agentic AI technologies and systems

Should such systems fail to accommodate ethical diversity, they risk reinforcing or exacerbating existing social and political inequalities. The uncritical deployment of AI in public contexts may entrench structural injustices, with far-reaching consequences for democratic legitimacy and social equity (Crawford, 2021).

5.4 Human-AI co-governance models

In light of these challenges, the governance of future agentic cities will likely necessitate the development of novel co-governance frameworks that integrate both human and non-human actors. Conventional regulatory paradigms, which regard technology as a passive instrument, are increasingly inadequate. Instead, governance models must recognise AI as an active and participatory agent within urban ecosystems and prioritise participatory design processes to ensure that technological systems support, rather than subvert, democratic urban futures.

Proposals such as the establishment of AI ombudspersons or algorithmic oversight councils build upon existing institutional precedents in the domain of data protection. These include roles such as data protection officers under the General Data Protection Regulation (GDPR) (European Union, 2016), algorithmic impact assessments employed in public sector services, such as those implemented by New York City's Automated Decision Systems Task Force (ADSTF, 2019), and human rights commissions tasked with identifying and addressing systemic bias. Such institutional analogues demonstrate that mechanisms for the oversight of complex and opaque systems can be operationalised through legally and ethically robust structures.

To this end, transparent audit mechanisms, participatory governance architectures, and rights-based frameworks for human–AI interaction will be essential to ensuring that agentic urban intelligences function within ethically grounded and democratically legitimate parameters.

Eventually, acknowledging the presence of non-human agency in urban environments does not necessitate the rejection of AI as a governance partner. Rather, it calls for the deliberate and critical construction of institutional frameworks through which human and artificial intelligences may collaboratively shape urban futures that are sustainable, equitable, and resilient.

To support the operationalisation of such governance responses, Table 3 outlines the principal challenges posed by Agentic Urban AI alongside proposed frameworks and strategies to address them.

6 A research agenda for agentic urban AI

Agentic AI constitutes the next significant frontier in the evolution of artificial intelligence, marking a transition from function-specific systems to context-aware, goal-oriented agents (Pounds, 2024). This advancement highlights the pressing need for a structured research agenda capable of addressing its implications and supporting the responsible development and integration of such systems within complex and dynamic urban environments. Building upon existing calls for more comprehensive investigation into autonomous urban systems (Cugurullo, 2020), this agenda foregrounds the emerging ethical, political, and agentic challenges that arise in the context of urban AI.

Comprehending the nature and implications of Agentic Urban AI necessitates the development of novel theoretical frameworks, alongside sustained interdisciplinary collaboration spanning urban studies, AI ethics, governance, law, sociology, and philosophy. Such an approach is essential to adequately address the conceptual and practical complexities associated with the rise of agentic systems and their transformative potential. The following section delineates key research priorities required to advance understanding, inform policy, and foster critical engagement with the evolving para-digm of the agentic city.

6.1 Mapping the emergence of urban AI agency

Understanding the empirical indicators of agency in urban AI aligns closely with recent calls to investigate how AI is materially and institutionally reshaping urban environments. Caprotti et al. (2024) identify this as a critical direction for future research, arguing that AI does not merely overlay smart systems but interacts dynamically with the city's infrastructures, institutions, and imaginaries. They call for in-depth studies of the interfaces through which AI "goes urban"—not only as technology, but as a new form of urban rationality.

Longitudinal investigations across a range of urban contexts will be vital in tracing the progression from autonomy to agency. Such studies should pay particular attention to discrepancies between pre-programmed objectives and emergent behaviours, as well as to instances wherein AI systems prioritise goals that have not been explicitly defined by human designers. Next generation sensing systems powered by LLMs are already capable of inferring urban intent from multi-modal data streams and may serve as critical indicators of emergent agency (Hou et al., 2025).

Table 3 Governance challenges vs. required frameworks for agentic Urban AI

Governance Challenge	Description	Required Frameworks/Solutions
Accountability and Attribution	Difficulty assigning responsibility for decisions made independently by AI agents	AI audit trails, AI ombudsperson roles, legal recognition of machine agency, transparent logs
Dynamic Goal Evolution	AI may change objectives over time, destabilizing policy continuity	Human-in-the-loop oversight, conditional override mechanisms, adaptive policy alignment protocols
Ethical Pluralism and Value Conflicts	Urban AIs may not reconcile competing human values or social equity concerns	Value-sensitive design, participatory governance models, embedded ethics modules in AI architecture
Transparency and Explainability	AI decision-making processes may be opaque ("black box") to stakeholders	Explainable AI (XAI), decision traceability, open-source models for public inspection
Human-AI Co-Governance	Integrating AI systems into democratic urban decision-making structures	Hybrid governance models, civic engagement platforms, AI-inclusive urban planning procedures

Source: Author, 2025

6.2 Developing ethical frameworks for urban AI agency

Ethical frameworks for agentic urban AI may draw upon established principles in AI ethics, which emphasise beneficence, justice, and explicability in the design and deployment of intelligent systems (Floridi & Cowls, 2022). However, the distinctive ethical challenges posed by agentic urban AI necessitate the formulation of novel frameworks that transcend conventional approaches to machine ethics. These frame-works must be capable of addressing the operational realities of multi-agent systems functioning within morally pluralistic and socially heterogeneous urban environments.

Several critical areas of inquiry emerge in this context:

How might agentic AI systems be designed to uphold human rights and demo-cratic principles?
What procedural mechanisms should be established to resolve conflicts between AI-generated goals and human political decisions?
In what ways can cities institutionalise ethical oversight of non-human urban ac-tors?

Accordingly, ethical frameworks for agentic urban AI must be highly con-text-sensitive, attuned to the plurality of values and interests that characterise con-temporary urban settings, as previously noted in discussions of smart city governance.

6.3 Innovating governance models for human-AI co-existence

The emergence of agentic cities necessitates the development of novel governance architectures capable of regulating hybrid polities comprising both human and non-human decision-makers. Future research should prioritise the exploration of hu-man–AI co-governance models, with particular attention to institutional structures that can facilitate accountable and equitable decision-making processes. Potential frameworks may include:

- The establishment of AI accountability boards embedded within municipal governance structures.
- Participatory governance models that synthesise machine-generated recommendations with human deliberation and oversight.
- Procedural mechanisms for human intervention, correction, or reprogramming of agentic AI systems under specified conditions.

The development and refinement of such models may be advanced through the use of simulations, participatory design workshops, and experimental governance trials, all of which can provide valuable insights into the operationalisation of co-governance in complex urban settings.

6.4 Ensuring value alignment and transparency

Another key research priority pertains to the mechanisms by which agentic urban AI systems may be aligned with evolving human values. Methods such as value learning, normative multi-agent negotiation, and explainable artificial intelligence (XAI) must be critically adapted and contextualised for application within urban governance frameworks.

In addition, research should investigate both technical and institutional strategies to promote transparency in AI decision-making. Such transparency must extend be-yond operational processes to encompass the prioritisation of values, the selection of goals, and the ethical reasoning underpinning AI behaviour. Ensuring transparency in these dimensions is fundamental to fostering and sustaining public trust in AI systems.

6.5 Scenario modelling and risk anticipation

Given the potential for significant unintended consequences, rigorous scenario modelling and anticipatory risk assessment are imperative. Researchers should develop models capable of exploring a range of possible trajectories for the development of agentic AI in urban contexts, from beneficial symbiosis to catastrophic misalignment of goals.

Such scenario planning ought to incorporate insights from complexity theory, urban political ecology, and technological risk analysis, in order to account for the un-predictable and emergent characteristics of urban systems influenced by agentic AI. Environmental modelling must also take into consideration the infrastructural demands of such systems, particularly in light of current concerns regarding the energy consumption of AI-driven data centres. Initiatives such as CyrusOne's solar-powered facility in Milan exemplify both the opportunities and the environmental risks associated with scaling agentic technologies without exacerbating ecological burdens (AI Agentic Store, 2025).

Nevertheless, significant warnings have been issued regarding the dangers posed by unregulated technological infrastructures that commodify human behaviour (Zuboff, 2019). Scenario models must therefore address not only the pursuit of operational efficiencies but also the potential emergence of power asymmetries, particularly those intensified through surveillance-oriented AI deployments in urban settings. A critical foresight agenda must attend to both technical resilience and ethical robustness, thereby ensuring that the integration of agentic AI into urban systems proceeds in a responsible and equitable manner.

6.6 Comparative and cross-cultural analysis

The development of agentic urban AI will not proceed uniformly across global contexts. Comparative research is essential to elucidate how differing political cultures, regulatory traditions, and societal attitudes towards technology influence the evolution of agentic cities. Examining cases across democratic, authoritarian, and hybrid governance regimes will provide critical insights into how diverse urban environments shape both the opportunities and the risks associated with non-human urban agency.

By articulating this research agenda, the present paper seeks to stimulate a critical and proactive scholarly engagement with the profound transformations already underway in urban systems. The agentic city is no longer a speculative construct of the distant future; it constitutes an emerging reality that warrants immediate and sustained intellectual attention.

Nonetheless, certain limitations of this conceptual enquiry must be acknowledged. The primarily theoretical orientation of the analysis may lead to speculative claims concerning the prospective capabilities of AI and its implications for governance. Moreover, the empirical evidence presently available is limited, and largely drawn from early-stage implementations. Future research should therefore prioritise broader empirical investigations and longitudinal case studies to validate and refine the theoretical propositions advanced herein, and to support the development of robust governance frameworks.

7 Conclusion: towards the Agentic City

Expanding upon the argument that Urban AI "transforms the smart-city imaginary" (Caprotti et al., 2024), this paper argues that Agentic Urban AI heralds a post-smart paradigm driven by urban sensing and LLMs (Hou et al., 2025), enabling dynamic goal setting, reshapes governance, ethics, and urban ecosystems as cities integrate biological and artificial intelligences.

Where smart-city programmes primarily pursued real-time optimisation, AI urbanism introduces open-ended autonomy and emergent socio-technical logics that recast the city as a site of distributed, non-human agency. Scholars emphasise that urbanistic solutions must address this shift, advocating governance frameworks that integrate AI's agency axis (Cugurullo et al., 2024). This transformation has far-reaching implications for governance, ethics, and the urban ecosystem itself, as cities increasingly integrate both biological and artificial intelligences in shaping their collective futures.

Agentic AI systems hold considerable promise in addressing complex urban challenges such as climate resilience, resource management, and social equity at unprecedented scales (Yigitcanlar & Cugurullo, 2020). However, the potential for misaligned objectives, governance drift, and ethical opacity underscores the imperative to fore-ground human agency and democratic values to ensure the equitable integration of AI within urban systems (Green, 2019).

Strategic foresight is essential in this regard. Policymakers must establish experimental 'sandbox cities' to rigorously examine the governance and ethical implications of agentic AI (Ghose, 2024). Cities that proactively engage with this paradigm may well position themselves at the forefront of global urban innovation.

Finally, the emergence of the Agentic City necessitates a fundamental reimagining of urbanism, compelling scholars, planners, and policymakers alike to shape its trajectory with vision, deliberation, and accountability. Recent industry analyses characterise agentic AI as a paradigmatic shift in the development of artificial intelligence, one that requires a comprehensive reconsideration of how intelligent systems engage with the complexity of real-world environments (Pounds, 2024). This recognition reinforces the urgent need to construct ethically grounded, technically robust, and participatory governance frameworks capable of guiding the ascent of agentic urban systems.

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